

Exercise 2

The data contained in the `chdage` dataset represent the measurements on 100 patients of two variables: the age expressed in years (`AGE`) and a binary variable (`CHD`) which assumes value 1 if the individual has a coronary heart disease and 0 otherwise.

- a) To investigate whether there is a relationship between the probability of having a coronary heart disease and the age of the individuals, a researcher fitted a generalized linear model (using the canonical link function) that produced the following output:

	Estimate	Std. Error	z value	Pr(> z)
Intercept	-5.3095	1.1337	-4.68	0.0000
age	0.1109	0.0241	4.610	0.0000

Null deviance:	136.66
Residual deviance:	107.35

- a1) Write the statistical model corresponding to such output (assumptions and model specification).
a2) Write the interpretation of the coefficient associated with the age variable.
a3) Write the system of hypotheses and perform a test to compare the fitted model with a model that includes only the intercept. Comment the result.

Q2 $\hat{\beta}_2 = 0.1109$ WHEN AGE INCREASE OF ONE UNIT (KEEPING EVERYTHING CONSTANT) WE INCREASE THE ODDS OF CORONARY HEART DISEASE OF 0.1109

Q3 FITTED MODEL VS NULL MODEL
 $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$

$w_{obs} \sim \chi^2_1$ $w_{obs} = \frac{2 \cdot \log\left(\frac{\hat{L}(\text{MODEL})}{\hat{L}(\text{NULL})}\right)}{2} = 2 \cdot \left\{ \hat{L}(\text{MODEL}) - \hat{L}(\text{NULL}) + \hat{L}(\text{SAT.}) - \hat{L}(\text{SAT.}) \right\}$
 $= D(\text{NULL}) - D(\text{MODEL}) = 136.66 - 107.35 = 29.31$

$\alpha^{obs} = P(W_1 \leq w_{obs}) = 1 - P(W_1 > w_{obs}) \approx 1 - 0.999 \approx 0$ SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.1; \alpha = 0.5; \alpha = 0.01$ I REJECT H_0 AT ALL CONFIDENCE LEVEL
→ I PREFER THE MODEL WITH AGE RATHER THAN THE MODEL WITH ONLY THE INTERCEPT. AGE IS A USEFUL FEATURE IN DETECTING CORONARY HEART DISEASE

- b) The researcher then wonders whether the age might have a quadratic effect and adds the corresponding covariate to the model. The fitted model produced the following output:

	Estimate	Std. Error	z value	Pr(> z)
Intercept	?	4.2901	-0.99	0.3229
age	?	0.1947	0.315	0.7527
age ²	0.0005	0.0021	?	?

Null deviance:	136.66
Residual deviance:	107.29

- b1) Write the statistical model corresponding to such output.
b2) Complete the missing values in the table.
b3) Write the system of hypotheses and perform a test to compare the fitted model with a model that includes only the intercept. Comment the result.
b4) Write the system of hypotheses and perform a test to evaluate which model is preferable between model (a) and (b). Comment the result.

Q1 $\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i} = \eta_i$ $Y_i \sim \text{NBernoulli}(\pi_i)$
 $\eta_i = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}$ $i = 1, \dots, 100$

Q2 ① $\begin{cases} H_0: \beta_1 = 0 \\ H_1: \beta_1 \neq 0 \end{cases}$
 $z_1^{obs} \sim N(0, 1)$ $z_1^{obs} = \frac{\hat{\beta}_1 - \beta_1}{\text{SE}(\hat{\beta}_1)} \rightarrow \hat{\beta}_1 = \frac{-0.99 \cdot 4.2901}{1.2301} = -4.247$

② $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$
 $z_2^{obs} \sim N(0, 1)$ $z_2^{obs} = \frac{\hat{\beta}_2 - \beta_2}{\text{SE}(\hat{\beta}_2)} \rightarrow \hat{\beta}_2 = \frac{0.315 \cdot 0.1947}{0.1947} = 0.061$

③ $\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 \neq 0 \end{cases}$
 $z_3^{obs} \sim N(0, 1)$ $z_3^{obs} = \frac{\hat{\beta}_3 - \beta_3}{\text{SE}(\hat{\beta}_3)} = \frac{0.0005 - 0}{0.0021} = 0.238$

④ $\alpha_3^{obs} = P(|Z_3| \geq |z_3^{obs}|) = 2(1 - \Phi(|z_3^{obs}|)) \approx 1 - 0.80 \approx 0$
 $\alpha_3^{obs} = P(|Z_3| \geq |z_3^{obs}|) = 2(1 - \Phi(|z_3^{obs}|)) > 0.10$ DO NOT REJECT H_0

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$$\begin{cases} H_0: \beta_2 = \beta_3 = 0 \\ H_1: \text{THO} \end{cases}$$

$$W^{obs} \sim \chi^2_2$$

$$W^{obs} = \log\left(\frac{\hat{\ell}(\text{Model})}{\hat{\ell}(\text{Null})}\right) = 2 \left\{ \hat{\ell}(\text{Model}) - \hat{\ell}(\text{Null}) + \hat{\ell}(\text{SAT}) - \hat{\ell}(\text{SAT}) \right\} =$$

$$= D(\text{Null}) - D(\text{Model}) = 136.66 - 107.29 = 29.37$$



$\alpha^{obs} = P(W_2 > W^{obs}) = 1 - P(W_2 \leq W^{obs}) \approx 1 - 0.9999 \approx 0$ SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.1; \alpha = 0.05; \alpha = 0.01$ I REJECT H_0 AT ALL CONFIDENCE LEVEL

I PREFER A MODEL WITH AGE AND AGE² AS FEATURES WITH RESPECT TO A MODEL THAT CONTAINS ONLY THE INTERCEPT

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$$\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 \neq 0 \end{cases}$$

$$W_3^{obs} \sim \chi^2_1$$

$$W_3^{obs} = \log\left(\frac{\hat{\ell}(\text{Model})}{\hat{\ell}(\text{REST.})}\right) = 2 \left\{ \hat{\ell}(\text{Model}) - \hat{\ell}(\text{REST.}) - \hat{\ell}(\text{SAT}) + \hat{\ell}(\text{SAT}) \right\} =$$

$$= D(\text{REST.}) - D(\text{Model}) = 107.35 - 107.29 = 0.06$$

$$\alpha^{obs} = P(W_1 > W^{obs}) = 1 - P(W_1 \leq W^{obs}) \approx 1 - 0.98 \approx 0.02$$



SINCE $\alpha^{obs} > \alpha$ FOR $\alpha = 0.1; \alpha = 0.05; \alpha = 0.01$ I DO NOT REJECT H_0 AT ALL CONFIDENCE LEVEL

I PREFER THE MODEL WITH ONLY AGE AND NOT WITH RESPECT TO A MODEL WITH BOTH AGE AND AGE²

$W^{obs} > \alpha \rightarrow \alpha^{obs} < \alpha$ REJ. H_0

$W^{obs} < \alpha \rightarrow \alpha^{obs} > \alpha$ NO REJ. H_0

c) To further investigate the relationship between the age and the presence of heart disease, the age variable was then transformed into a dummy variable. Specifically, the new variable age<50 takes value 1 if age is smaller than 50 and 0 otherwise. With this new variable, the following output is produced when fitting the model:

	Estimate	Std. Error	z value	Pr(> z)
Intercept	1.0609	0.3867	2.74	0.0061
age<50	-2.0989	0.4788	-4.38	0.0000

Null deviance:	136.66
Residual deviance:	114.61

- c1) Write the statistical model corresponding to such output.
c2) Write the interpretation of the slope coefficient.

X1

$$M_i = \beta_1 + \beta_2 D_i$$

$$Y_i \sim \text{BERNOULLI}(\pi_i) \quad i = 1, \dots, 100 \quad n = 100$$

$$D_i = \begin{cases} 1 & \text{if AGE OF } i \text{ IS SMALLER THAN 50} \\ 0 & \text{OTHERWISE} \end{cases}$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_1 + \beta_2 D_i \quad ; \quad M_i = g(\pi_i)$$

X2

$$\hat{\beta}_2 = -2.0989$$

(WHEN KEEPING EVERYTHING CONSTANT) IF WE MOVE AGE FROM SMALLER THAN 50 TO ABOVE 50 WE INCREASE THE ODDS OF HEART DISEASE BY 2.0989

THE LOG ODDS OF AN INDIVIDUAL ABOVE THE AGE OF 50 TO GET CARDIOVASCULAR DISEASE ARE EQUAL TO THE ONE OF AN INDIVIDUAL BELOW THAN 50 MULTIPLIED BY -2.0989